

# Hemanth A R

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## PROFESSIONAL SUMMARY

Final-year Information Science Engineering student with hands-on experience across AI, cybersecurity, backend, and full-stack software engineering. Designed and developed production-oriented software spanning an AI-powered cybersecurity investigation platform, a local-first AI knowledge platform, and a Zero Trust Network Access (ZTNA) solution. Led the development of a Zero Trust Network Access (ZTNA) final-year project, strengthening expertise in secure system design and identity-based access control. Strong foundation in software architecture, Retrieval-Augmented Generation (RAG), secure application development, and scalable backend systems.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, SQL

**Backend:** Flask, Node.js, Express.js, REST APIs, JWT Authentication

**Frontend:** React, HTML5, CSS3, Tailwind CSS, Vite

**AI:** Retrieval-Augmented Generation (RAG), Google Gemini API, Ollama, SentenceTransformers, Vector Search

**Cybersecurity:** MITRE ATT&CK, Threat Intelligence, Zero Trust (ZTNA), RBAC, OAuth 2.0, OpenID Connect (OIDC), Keycloak

**Databases:** SQLite, MongoDB, ChromaDB, LanceDB, PostgreSQL

**Tools:** Git, GitHub, Docker, Docker Compose, Nginx, Postman

## PROFESSIONAL EXPERIENCE

### Full Stack Development Intern

Feb 2026 – May 2026

*VisionAstra EV Academy*

*Bengaluru, India*

- Developed the EV Service & Vehicle Management System, a MERN web application supporting authentication, vehicle management, EV service booking, maintenance tracking, and administrative operations.
- Engineered RESTful APIs with Node.js, Express.js, and MongoDB, and built responsive React interfaces integrating CRUD operations and authentication.
- Applied Git, Postman, debugging, software testing, and technical documentation across the development lifecycle.

## PROJECTS

### AI Personal OS | *Local-First AI Knowledge Platform* | [GitHub](#)

- Independently architected and developed a local-first AI platform that transforms personal documents into structured, searchable knowledge while preserving complete user privacy through offline processing.
- Engineered an offline ingestion pipeline with OCR fallback, SHA-256 deduplication, deterministic chunking, local embeddings, and LanceDB vector indexing.
- Established the technical foundation by authoring a PRD, 16 Architecture Decision Records (ADRs), design documentation, and milestone-driven implementation plans ahead of development.

### SecureRAG | *AI-Powered Cybersecurity Investigation Platform* | [GitHub](#)

- Engineered an AI-powered cybersecurity investigation platform using Retrieval-Augmented Generation (RAG), IOC correlation, MITRE ATT&CK mapping, and evidence-based AI responses.
- Built a modular architecture spanning 24 REST endpoints, JWT authentication, RBAC, ChromaDB semantic retrieval, SQLite SIEM storage, Docker deployment, and CI workflows.
- Improved retrieval quality by increasing Recall@1 from 0.80 to 0.90 and MRR from 0.90 to 0.95 using cross-encoder reranking.

### Zero Trust Network Access (ZTNA) | *Final-Year Group Project · Team Lead* | *Keycloak, OAuth 2.0, Docker*

- Led the final-year project team in designing and implementing a Zero Trust Network Access platform using Keycloak, OAuth 2.0, OpenID Connect, PostgreSQL, and Docker.
- Designed secure identity and access workflows implementing SSO, RBAC, and least-privilege authorization.

## EDUCATION

### Cambridge Institute of Technology

Bengaluru, India

*Bachelor of Engineering (Information Science & Engineering); CGPA: 7.0/10*

*2022 – 2026*

## LANGUAGES

English · Kannada · Hindi